Supplementary Appendix:
Simulation-Based Verification of Return-Level Transformations, Bias, RMSE, and Convergence

Supplementary material for
Catch Me if You Can: Comparing Extreme Speed and Athleticism of NFL Cornerback and Wide Receiver Prospects

Integration Note

This document provides supplementary material for the manuscript *"Catch Me if You Can: Comparing Extreme Speed and Athleticism of NFL Cornerback and Wide Receiver Prospects."* The appendix is written to be fully standalone and may be incorporated verbatim as an additional appendix in the main manuscript.

1. Overview

This supplementary appendix presents a simulation-based investigation supporting the modeling and inferential choices employed in the main analysis. The simulation study is designed to: (i) validate the use of annual block minima of 40-yard dash times as extreme observations, motivated by classical extreme-value arguments and supported by empirical convergence diagnostics (Sections 3–5); (ii) assess the stability of the time-to-speed transformation when applied at the return-level stage via comparative bias and RMSE behavior across scales, with theoretical justification provided by the Continuous Mapping Theorem and the Delta Method (Sections 5–6); and (iii) characterize the finite-sample behavior of GEV-based return-level estimators as record length increases using Monte Carlo sampling distributions, empirical bias and RMSE, and estimated convergence rates (Sections 4–5).

The analysis is conducted on two closely related scales:

- the negative-time scale $X = -T$, under which faster performances correspond to larger values; and
- the speed scale $V = 40/T$ (yards per second).

For each scale, convergence is examined through Monte Carlo sampling distributions and summarized using boxplots, empirical bias, and root mean squared error (RMSE). The resulting empirical patterns are interpreted using standard probabilistic arguments, including convergence in probability, the Continuous Mapping Theorem, and the Delta Method.

2. Block Extremes and Scale Transformations

Let $T_{y,1}, \ldots, T_{y,n_y}$ denote observed 40-yard dash times (in seconds) recorded in year y, and define the annual fastest time by

$$T_y^{\min} = \min_{1 \le j \le n_y} T_{y,j}.$$

Annual extremes may be represented equivalently on two scales. On the negative-time scale,

$$M_y^{(-T)} = \max_{1 \le j \le n_y} (-T_{y,j}) = -T_y^{\min},$$

while on the speed scale,

$$M_y^{(V)} = \max_{1 \le j \le n_y} \left(\frac{40}{T_{y,j}} \right) = \frac{40}{T_y^{\min}}.$$

In both cases, annual extremes correspond to the fastest observed performances, expressed either as negative time or as speed. Because the transformation between the two scales is strictly monotone, both representations induce the same ordering of extremes and encode equivalent extremal information. This formulation aligns with the classical block maxima framework in extreme-value theory, in which one extreme observation is retained per block and modeled using generalized extreme value limits.

3. Simulation Design

3.1 Parent model and sampling scheme

To mimic annual samples of 40-yard dash times, each Monte Carlo replicate begins by generating independent and identically distributed observations from a parent model for T.[1] For example, the R implementation uses a normal distribution calibrated using an empirical mean and standard deviation:

$$T \sim \mathcal{N}(\mu_T, \sigma_T^2).$$

Although both wide receivers and cornerbacks are analyzed in the main manuscript, the simulation design is calibrated using wide receiver (WR) performance data as a representative benchmark. The theoretical arguments and finite-sample convergence properties examined in this appendix are not position-specific and are expected to extend to comparable athletic performance distributions.

The use of annual block minima is motivated by classical arguments from extreme-value theory. For each year, the fastest observed time corresponds to the minimum of a large collection of approximately i.i.d. performance times. Under mild regularity conditions, such block minima—after sign reversal—converge in distribution to a generalized extreme value (GEV) limit. This provides theoretical justification for treating annual fastest times as extreme observations and modeling them using GEV-based methods for return-level inference.

For a chosen record length (number of years) n and per-year sample size m, we simulate $n \times m$ realizations of T, partition them into n yearly blocks, and compute the annual minima $T_1^{\min}, \ldots, T_n^{\min}$. The corresponding

[1] In the simulations, the parent distribution parameters are calibrated using empirical summaries of wide receiver 40-yard dash times. For strict reproducibility across computing environments, the Monte Carlo study can be run with fixed calibration values reported in the manuscript (as implemented by the `freeze_parent` option in the accompanying R code). The qualitative convergence behavior examined here follows from general extreme-value and continuous-mapping arguments and does not depend on the exact numerical calibration.

block-maxima series on the negative-time scale is then defined as

$$X_i = -T_i^{\min}, \qquad i = 1, \ldots, n.$$

3.2 GEV fitting on $-T$ block maxima

For each Monte Carlo replicate, a generalized extreme value (GEV) model is fitted to the block-maxima series $\{X_i\}_{i=1}^n$ on the negative-time scale, yielding parameter estimates $\widehat{\mu}$ (location), $\widehat{\sigma}$ (scale), and $\widehat{\xi}$ (shape). For return periods p of interest, the corresponding return levels are computed on the $-T$ scale and denoted $\widehat{\mathrm{RL}}^{(-T)}(p)$.

3.3 Speed-scale return levels via transformation

Rather than fitting a separate GEV model directly to the annual speed extremes $\{M_y^{(V)}\}$ in the simulation, return levels on the speed scale are obtained by transforming the estimated return levels on the negative-time scale. Specifically, for a return period p, the speed-scale return level is defined as

$$\widehat{\mathrm{RL}}^{(V)}(p) = \frac{40}{-\widehat{\mathrm{RL}}^{(-T)}(p)}.$$

This procedure applies the same reciprocal mapping used to define speed from time, but does so at the return-level stage rather than at the level of the block extremes themselves. As a result, both the negative-time and speed-scale return levels are derived from a common fitted extremal model, enabling a direct comparison of finite-sample behavior across scales and an explicit assessment of the stability of the time-to-speed transformation.

3.4 Monte Carlo setup

Throughout the simulation study, $R = 1{,}000$ Monte Carlo replicates are used for each record length n. Record lengths are taken from the grid

$$n \in \{5, 10, 15, 20, 25, 26, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100\},$$

spanning short to long samples and including values motivated by the empirical structure of the observed data.

For each n in this grid, the following procedure is carried out:

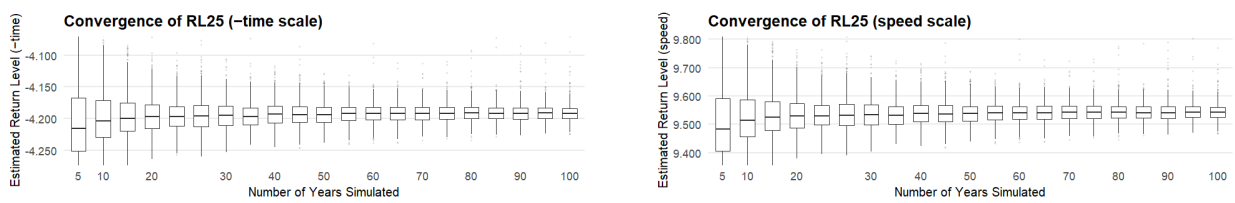
1. Generate n annual block minima from independent draws of T and form the corresponding negative-time extremes $X_i = -T_i^{\min}$, $i = 1, \ldots, n$;
2. Fit a GEV model to $\{X_i\}_{i=1}^n$ and compute return-level estimates $\widehat{\mathrm{RL}}^{(-T)}(p)$ for the return periods of interest;
3. Obtain speed-scale return levels by applying the reciprocal transformation $\widehat{\mathrm{RL}}^{(V)}(p) = 40/(-\widehat{\mathrm{RL}}^{(-T)}(p))$;
4. Repeat steps (1)–(3) independently over R replicates to produce Monte Carlo sampling distributions as a function of n.

4. Convergence Diagnostics via Boxplots

Convergence is examined by assessing how the Monte Carlo sampling distributions of $\widehat{\mathrm{RL}}^{(-T)}(p)$ and $\widehat{\mathrm{RL}}^{(V)}(p)$ evolve as the record length n increases. Boxplots across replicates provide a compact summary of dispersion, central tendency, and stability of the estimators at each n. As n grows, increasing concentration of the sampling distributions reflects improved estimator precision and provides empirical evidence consistent with asymptotic convergence.

In some settings, a small number of extreme Monte Carlo realizations can dominate the vertical scale of the boxplots, obscuring the behavior of the bulk of the sampling distribution. In such cases, boxplots restricted to central quantiles are used as a visual aid to highlight typical estimator behavior and to facilitate comparison across record lengths. This restriction is used solely for visualization; all simulations are retained in the computation of empirical bias, RMSE, and convergence diagnostics.

Boxplots for additional return periods and for both scales exhibit the same qualitative tightening and stabilization patterns and are therefore omitted for brevity.



(a) $-T$ scale (negative time). **(b)** Speed scale $40/T$.

Figure 1. Monte Carlo boxplots of the estimated 25-year return level, $\widehat{\mathrm{RL}}(25)$, as a function of record length n, shown on the negative-time and speed scales.

5. Empirical Bias and RMSE

5.1 Reference "truth" via long-run simulation

To benchmark empirical bias and root mean squared error (RMSE) in settings where closed-form reference values are unavailable, a long-run simulation with a large record length n_{true} is used to construct high-fidelity numerical reference quantities. Specifically, we obtain

$$(\mu^\star, \sigma^\star, \xi^\star), \qquad \mathrm{RL}^{(-T),\star}(p), \qquad \mathrm{RL}^{(V),\star}(p) = \frac{40}{-\mathrm{RL}^{(-T),\star}(p)}.$$

These quantities serve as reference values for evaluating finite-sample bias and RMSE across record lengths. In practice, these reference quantities are obtained using a long-run simulation with record length $n_{\text{true}} = 100{,}000$, which is sufficiently large to make Monte Carlo error negligible relative to the finite-sample effects of interest.

5.2 Definitions

For a given return period p and record length n, let $\widehat{\mathrm{RL}}_r^{(s)}(p; n)$ denote the return-level estimate obtained from Monte Carlo replicate $r = 1, \ldots, R$ on scale $s \in \{-T, V\}$. Empirical bias and root mean squared error

(RMSE) are defined as

$$\widehat{\mathrm{Bias}}^{(s)}(p;n) = \frac{1}{R}\sum_{r=1}^{R}\left(\widehat{\mathrm{RL}}_r^{(s)}(p;n) - \mathrm{RL}^{(s),\star}(p)\right),$$

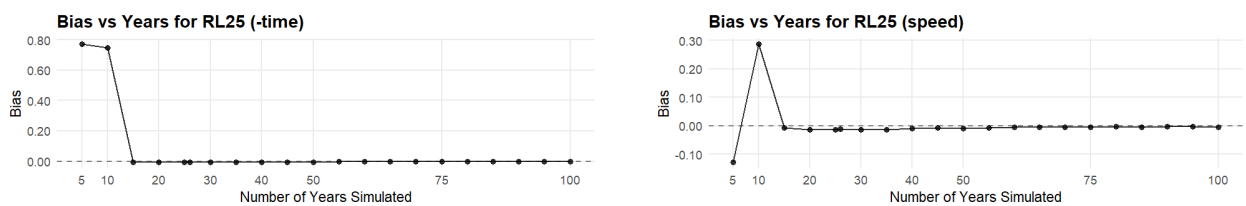
$$\widehat{\mathrm{RMSE}}^{(s)}(p;n) = \left[\frac{1}{R}\sum_{r=1}^{R}\left(\widehat{\mathrm{RL}}_r^{(s)}(p;n) - \mathrm{RL}^{(s),\star}(p)\right)^2\right]^{1/2}.$$

These quantities summarize systematic bias and overall estimation error as functions of record length n.

5.3 Bias and RMSE for $\mathrm{RL}(25)$ on both scales

To illustrate finite-sample performance under the proposed modeling framework, we examine empirical bias and RMSE for the 25-year return level on both the negative-time and speed scales.

Figure 2 displays empirical bias as a function of record length. On both scales, bias decreases toward zero as n increases, consistent with consistency of the GEV-based return-level estimator. At smaller record lengths, bias is often modestly larger on the speed scale, reflecting the nonlinearity of the reciprocal transformation applied at the return-level stage; the discrepancy attenuates with increasing n.

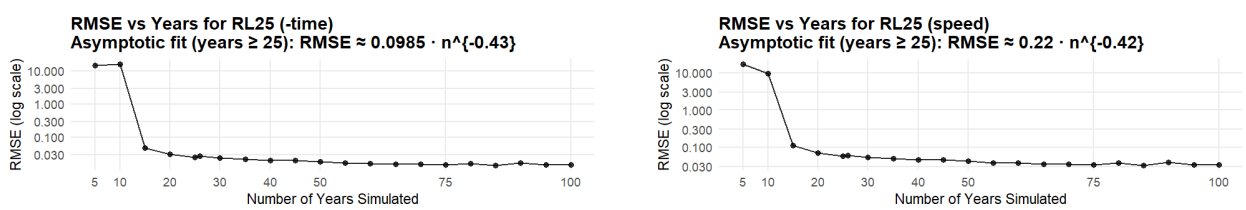


(a) Bias for $\widehat{\mathrm{RL}}^{(-T)}(25)$. **(b)** Bias for $\widehat{\mathrm{RL}}^{(V)}(25)$.

Figure 2. Empirical bias versus record length n for $\widehat{\mathrm{RL}}(25)$ on the $-T$ scale (left) and speed scale (right).

Figure 3 summarizes RMSE behavior on a logarithmic scale. RMSE decreases monotonically with increasing record length on both scales, and the approximately linear decay on the log–log scale suggests a power-law relationship between RMSE and n, consistent with asymptotic root-n convergence.

Across record lengths considered, RMSE is uniformly larger on the speed scale than on the negative-time scale. The decay patterns are nevertheless nearly parallel across scales, indicating that the time-to-speed transformation primarily inflates variability while preserving the underlying convergence rate.



(a) RMSE for $\widehat{\mathrm{RL}}^{(-T)}(25)$ (log scale). **(b)** RMSE for $\widehat{\mathrm{RL}}^{(V)}(25)$ (log scale).

Figure 3. Empirical RMSE versus record length n for $\widehat{\mathrm{RL}}(25)$ on the $-T$ scale (left) and speed scale (right).

5.4 Empirical convergence-rate estimation and theory comparison

To quantify the rate at which return-level estimators stabilize as record length increases, we estimate empirical convergence rates for each return period p by fitting a log–log regression to the RMSE curves over an approximately asymptotic range of record lengths. Specifically, we fit

$$\log\Big(\widehat{\mathrm{RMSE}}^{(s)}(p;n)\Big) \approx \log C^{(s)}(p) - \alpha^{(s)}(p)\log n,$$

where $s \in \{-T, V\}$ denotes the negative-time and speed scales, respectively. The fitted slope $\alpha^{(s)}(p)$ represents the empirical decay exponent, while $C^{(s)}(p)$ and the coefficient of determination R^2 summarize the scale and goodness-of-fit of the power-law approximation.

A natural theoretical benchmark is $\alpha \approx 1/2$, corresponding to root-n convergence. In addition, when speed-scale return levels are obtained via the smooth transformation $g(t) = 40/t$ applied to the negative-time return levels, the Delta Method predicts the relative dispersion across scales. Evaluating g near $T^\star = -\mathrm{RL}^{(-T),\star}(p)$ yields

$$\frac{\mathrm{RMSE}^{(V)}(p;n)}{\mathrm{RMSE}^{(-T)}(p;n)} \approx |g'(T^\star)| = \frac{40}{(T^\star)^2}.$$

This theoretical inflation factor is compared with the empirically estimated ratio $C^{(V)}(p)/C^{(-T)}(p)$ obtained from the fitted power-law models.

Table 1 reports the estimated decay exponents, scale parameters, and goodness-of-fit measures for both scales across return periods. The results show that empirical convergence rates on the negative-time and speed scales are closely aligned, and the magnitude inflation induced by the reciprocal transformation is well approximated by the Delta Method.

6. Theoretical Justification

6.1 Markov inequality and convergence in probability

For any $\varepsilon > 0$, Markov's inequality applied to the squared error $(\widehat{\theta} - \theta)^2$ yields

$$\mathbb{P}\big(|\widehat{\theta} - \theta| \geq \varepsilon\big) \leq \frac{\mathbb{E}\big[(\widehat{\theta} - \theta)^2\big]}{\varepsilon^2}.$$

Consequently, if $\mathbb{E}\big[(\widehat{\theta} - \theta)^2\big] \to 0$ as $n \to \infty$, then $\widehat{\theta} \xrightarrow{p} \theta$. Thus, decreasing RMSE with increasing record length is consistent with convergence in probability of the estimator under squared-error loss.

6.2 Continuous Mapping Theorem (reciprocal transformation)

If $T_n \xrightarrow{p} T$ with $T > 0$, then for the continuous mapping $g(t) = 40/t$ defined on $(0, \infty)$,

$$\frac{40}{T_n} = g(T_n) \xrightarrow{p} g(T) = \frac{40}{T}.$$

This provides formal justification for stability of the time-to-speed transformation at the return-level stage: consistency on the negative-time scale is preserved under the reciprocal map, provided the limiting return level remains strictly positive.

Table 1. Empirical power-law fits for RMSE decay and transformation-based comparisons.

p	$-T$ scale			Speed scale $V = 40/T$			Transformation comparison	
p	α_{-T}	C_{-T}	R^2_{-T}	α_V	C_V	R^2_V	C_V/C_{-T}	$\lvert g'(T^\star)\rvert$
5	-0.047	0.011	0.025	-0.066	0.023	0.045	2.091	2.220
10	0.159	0.026	0.312	0.144	0.056	0.257	2.135	2.247
15	0.290	0.049	0.700	0.279	0.106	0.666	2.175	2.262
20	0.372	0.073	0.848	0.364	0.162	0.831	2.207	2.272
25	0.426	0.098	0.908	0.421	0.219	0.898	2.234	2.279
26	0.435	0.103	0.915	0.430	0.230	0.906	2.239	2.280
30	0.465	0.122	0.936	0.461	0.276	0.929	2.258	2.285
35	0.494	0.145	0.951	0.492	0.331	0.945	2.280	2.290
40	0.517	0.167	0.959	0.517	0.384	0.955	2.300	2.294
45	0.535	0.188	0.964	0.537	0.436	0.960	2.320	2.298
50	0.551	0.208	0.967	0.554	0.486	0.963	2.339	2.301
55	0.564	0.227	0.969	0.569	0.536	0.964	2.358	2.304
60	0.576	0.246	0.971	0.582	0.584	0.965	2.378	2.307
65	0.586	0.264	0.971	0.594	0.632	0.965	2.397	2.309
70	0.595	0.281	0.971	0.605	0.679	0.965	2.416	2.311
75	0.603	0.298	0.971	0.615	0.726	0.964	2.436	2.313
80	0.611	0.315	0.971	0.625	0.773	0.963	2.456	2.315
85	0.618	0.331	0.971	0.633	0.820	0.961	2.476	2.317
90	0.625	0.347	0.970	0.642	0.867	0.960	2.497	2.318
95	0.631	0.363	0.970	0.650	0.914	0.958	2.518	2.320
100	0.636	0.378	0.969	0.657	0.961	0.957	2.540	2.321

6.3 Delta Method (variance propagation under transformation)

If $\sqrt{n}(T_n - T) \Rightarrow \mathcal{N}(0, \sigma_T^2)$ and g is differentiable at T, then the Delta Method implies

$$\sqrt{n}\big(g(T_n) - g(T)\big) \Rightarrow \mathcal{N}\Big(0,\ (g'(T))^2\sigma_T^2\Big).$$

For $g(t) = 40/t$, we have $g'(T) = -40/T^2$, yielding the variance approximation

$$\mathrm{Var}\left(\frac{40}{T_n}\right) \approx \left(\frac{40}{T^2}\right)^2 \mathrm{Var}(T_n).$$

This relationship clarifies how dispersion propagates under the time-to-speed transformation and provides a theoretical basis for the scale-inflation benchmarks reported in Table 1.

More generally, under standard regularity conditions for maximum likelihood estimation, the estimator of the GEV parameter vector $\theta = (\mu, \sigma, \xi)$ satisfies

$$\sqrt{n}(\widehat{\theta} - \theta) \Rightarrow \mathcal{N}\Big(0, I_E(\theta)^{-1}\Big),$$

where $I_E(\theta)$ denotes the expected Fisher information matrix. Since return levels are smooth functions of θ, a

further application of the Delta Method yields

$$\sqrt{n}\big(\widehat{\mathrm{RL}}(p) - \mathrm{RL}(p)\big) = O_p(1),$$

so the RMSE of the return-level estimator decays at rate $n^{-1/2}$, providing a natural theoretical benchmark for the empirical convergence exponents estimated in Section 5.

Reproducibility

All simulation results were generated using R and the `extRemes` package. The simulation framework includes Monte Carlo generation of annual block minima, maximum likelihood estimation of GEV models on the negative-time scale, return-level computation and transformation to the speed scale, and empirical evaluation of bias, RMSE, and convergence rates. Complete reproducible code is available from the authors upon request.